

EECS 2311
 LAB 5 (Take Home Assignment)
 Otokpa Attah - 220371068

USER STORY	MAIN DEVELOPERS	MEMBER TESTING THE IMPLEMENTATION
Each Tab Functional (ITR1) Allow the user to swap between tabs during their session. These tabs do not have to be fully functional yet but being to swap between them must be. Priority Level: High, Cost: 4 days	Otokpa Attah Pietro Angiolella Rishab Yadav	Mohammad Waliduddin
Add User Authentication (ITR1) Enable users to create accounts, log in, and persist data across sessions. Priority Level: High, Cost: 3 days	Rishab Yadav	Pietro Angiolella
Store User Search History (ITR1) Store the user's search history, allowing users to revisit previous topics they've searched for. Priority Level: High, Cost: 2 days	Rishab Yadav Ondrea Peters	Pietro Angiolella
Display External Resources (ITR1) For each search, display at least 1 external resource or video. Priority Level: High, Cost: 3 days	Mohammad Waliduddin Rishab Yadav	Larissa Singh
Search Course Functional (ITR2) Be able to search a course and for it to grab the Rate Your Prof ratings for the next semester and a quick overview. Priority Level: High, Cost: 5 days	Rishab Yadav Otokpa Attah Larissa Singh Mohammad Waliduddin	Ondrea Peters
Refine Search Topic (ITR2) Refine the search process and roadmap generation. Limit google searches, better search options and be able to click on each Roadmap topic. Priority Level: High, Cost: 3 days	Pietro Angiolella Rishab Yadav	Otokpa Attah
Quiz Generation Possible (ITR2) Have a semi-working quiz generation tab. Will be refined in the coming iterations. Priority Level: Medium, Cost: 3 days	Pietro Angiolella	Rishab Yadav
Refined History (ITR2) Have a working history system with a favouriting system. Priority Level: Medium, Cost: 3 days	Pietro Angiolella Rishab Yadav	Rishab Yadav

User Story	Problem Number	Description	Can reproduce	Severity
Search Course	1	Unrelated	Yes	Medium

Functionality (ITR 2)		school topics can be searched, with a video and learning roadmap		
Search Course Functionality (ITR 2)	2	Roadmap produces error on occasion when searching topics	Sometimes	High

1) TASK 1

Test Case 1 (Basic Search):

- Steps:
 - Open "Search" tab
 - Type "Chemistry"
 - Click Search
- Expected:
 - Videos related to chemistry appear
 - Roadmap steps appear related to chemistry
- Actual:
 - Videos related to chemistry appear
 - Roadmap steps appear related to chemistry

Test Case 2 (Nothing in the search query):

- Steps:
 - Open "Search" tab
 - Type nothing
 - Click search
- Expected:
 - Nothing appears
- Actual:
 - Nothing appears

Test Case 3 (Signs in the search query):

- Steps:
 - Open "Search" tab
 - Type "Chemistry@!"
- Expected:
 - Videos related to chemistry appear
 - Roadmap steps appear related to chemistry
- Actual:
 - Videos related to chemistry appear
 - Roadmap steps appear related to chemistry

Test Case 4 (Long words searched in query):

- Steps:
 - Open "Search" tab
 - Type long essay (*lorem ipsum* in my test case)
- Expected:
 - No video results are showed
 - Some roadmap results are shown according to some words from the essay
- Actual:
 - No video results are showed
 - Some roadmap results are shown according to some words from the essay

Test Case 5 (Changing number of roadmap steps before search):

- Steps:
 - Open "Search" tab
 - Type topic in search bar
 - Change the roadmap step
- Expected:
 - Video results related to the topic appear
 - The number of roadmap results are equal to the amount selected prior
- Actual:
 - Video results related to the topic appear
 - The number of roadmap results are equal to the amount selected prior

Test Case 6 (Changing number of roadmap steps after search):

- Steps:
 - Open "Search" tab
 - Type topic in search bar
 - Click Search
 - Change the roadmap step
 - Click Search again
- Expected:
 - Video results related to the topic appear
 - The number of roadmap results are equal to the amount selected prior
- Actual:
 - Video results related to the topic appear
 - The number of roadmap results are equal to the amount selected prior (or returns error)

2) TASK 2

Code Smells

- a. Magic Numbers (Minor)
 - i. **Location:** `animateResults() delays (0.1 + (index * 0.2))s`.
 - ii. **Smell:** Hardcoded timing values make adjustments difficult
 - iii. **Fix:** Add more variables with the timing numbers
- b. Duplicated Reflow Trigger (Moderate)

- i. **Location:** animateResults() and animateStepSelector() both use:
`element.style.animation = 'none';`
`element.offsetHeight;`
- ii. **Smell:** Copy-pasted reflow logic violated DRY
- iii. **Fix:** Create a helper function
- c. Overly Specific Animations (Moderate)
 - i. **Location:** animateResults() containing slideUp/slideInRight
 - ii. **Smell:** Tight coupling between animation logic and DOM structure
 - iii. **Fix:** Use data attributes instead

BUG REPORT

- a. Magic Numbers In Timing
 - i. **File:** search-topic.js
 - ii. **Issue:** Hardcoded Delays
 - iii. **Can reproduce:** Yes
 - iv. **Severity:** Low
 - v. **Fix:**
`const BASE_DELAY = 0.1;`
`const DELAY_INCREMENT = 0.2;`
`panel.style.animationDelay = `${BASE_DELAY + (index * DELAY_INCREMENT)}s`;`
- b. Duplicated Reflow Logic
 - i. **File:** search-topic.js
 - ii. **Issue:** Repeated element.offsetHeight reflow triggers
 - iii. **Can reproduce:** No
 - iv. **Severity:** Low
 - v. **Fix:**
`function resetAnimation(element) {`
`element.style.animation = 'none';`
`void element.offsetHeight;}`
- c. Tightly Coupled Animation
 - i. **File:** search-topic.js
 - ii. **Issue:** Hardcoded slideUp/slideInRight animations
 - iii. **Can reproduce:** Yes
 - iv. **Severity:** Moderate
 - v. **Fix:** Use CSS to code animation